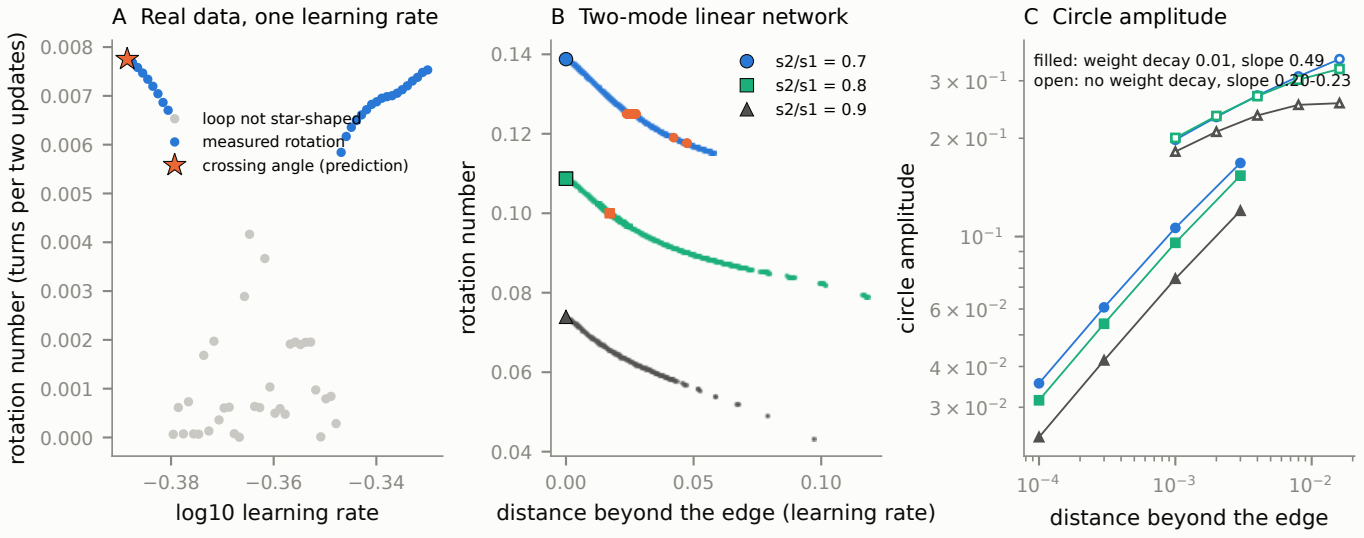
A Real data, one learning rate
rotation number (turns per two updates)
0.008
0.007
0.006
0.005
0.004
0.003
0.002
0.001
0.000
loop not star-shaped
measured rotation
crossing angle (prediction)
−0.38
−0.36
−0.34
log10 learning rate
B Two-mode linear network
rotation number
0.14
0.12
0.10
0.08
0.06
0.04
s2/s1 = 0.7
s2/s1 = 0.8
s2/s1 = 0.9
0.00
0.05
0.10
distance beyond the edge (learning rate)
C Circle amplitude
filled: weight decay 0.01, slope 0.49
open: no weight decay, slope 0.20-0.23
circle amplitude
3 × 10⁻¹
2 × 10⁻¹
10⁻¹
6 × 10⁻²
4 × 10⁻²
3 × 10⁻²
10⁻⁴
10⁻³
10⁻²
distance beyond the edge